Energy Conversion Laboratory Report 7

GROUP 11

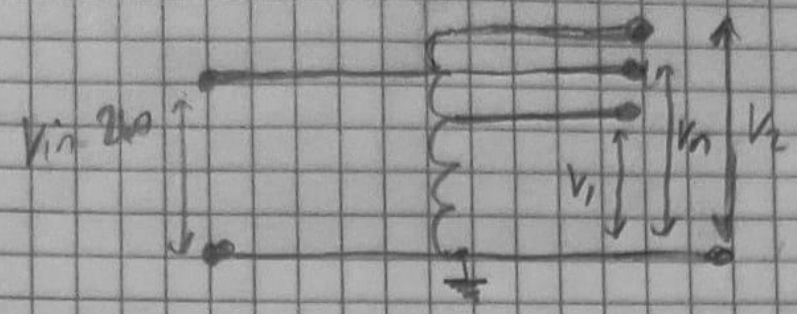
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Q1



Nominal Voltage = 240 V Tapping Range = ±5%

The turns Ratio = 0.9

$V_{out} = V_{in} \times 0.9$

$V_{out} = 240 \times 0.9$

$V_{out} = 216\,V = V_n$

Minimum voltage

$V_{min} = 240 - (0.05 \times 240)$

$V_{min} = 228\,V = V_1$

Maximum Voltage

$V_{max} = 240 + (0.05 \times 240)$

$V_{max} = 252\,V = V_2$

2. A 26 KVA, 260/130 V single phase transformer has an efficiency of 94% and a full load unity power factor. It is connected as an autotransformer with an input voltage of 260 V to feed a load of 390 V. What will the autotransformer's full load efficiency be at unity power factor? Draw the autotransformer circuit.

- Votage rating of auto transformer = 260 V / 390 V

K = 390/260 = 1,5

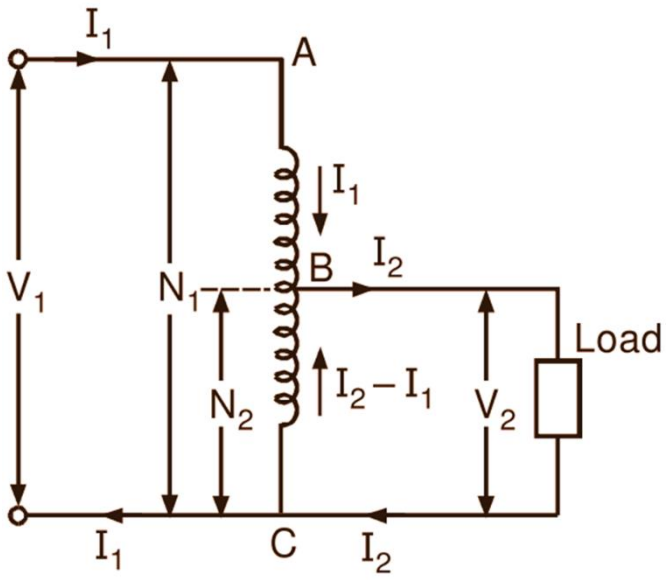
KVA rating of Auto transformer = (K / K-1) KVA rating of two winding transformer = 78 KVA

Total losses in 2 winding transformers = total losses in auto transformer

$$\eta_{2winding} = \frac{26 \cdot 10^3 \cdot 1}{26 \cdot 10^3 \cdot 1 + total\ losses} = 0.94$$

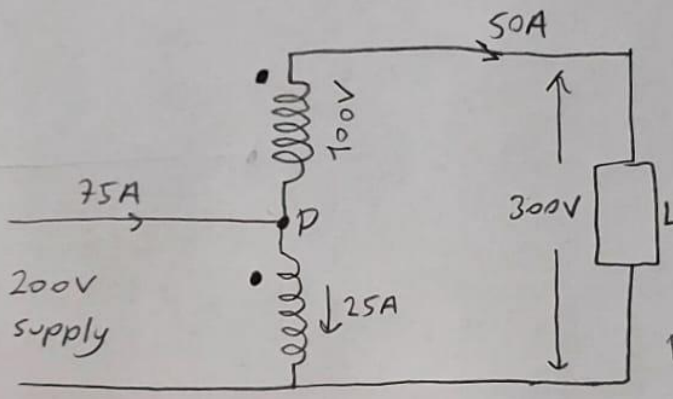
Total losses = 3,191 KW

$$\boldsymbol{\eta_{AutoT/F}} = \frac{78 \cdot 10^3 \cdot 1}{78 \cdot 10^3 \cdot 1 + 3{,}191 \cdot 10^3} = \mathbf{0,9607} = \mathbf{96,07}\ \%$$



Question - 3

Knowing the dot markings between the two coils is crucial to connect a two winding transformer as an autotransformer. Since the input voltage is 200 V, the supply must be connected to the HV terminals. The induced voltage on the LV side is fixed at 100 V, but to obtain a 300V output, the LV coil should be connected in series with the HV coil in an additive manner.



Rated voltage of HV coil = 200V

Rated voltage of LV coil = 100V

Phase turns ratio = $200/100 = 2$

Rated current of each HV coil = $\frac{5000}{200} = 25A$

Rated current of each LV coil = $\frac{5000}{100} = 50A$

Output kVA = $300 \times 50 = 15$ kVA

input kVA = Output kVA = 15 kVA

$\because$ transformer is ideal

Current drawn form the supply = $\frac{15000}{200} = 75A$

Current through HV coil $I_{HV} = 75 - 50 = 25A$

$\therefore$ kVA transferred magnetically = kVA of either HV or LV coil

$= 200 \times 25VA = 100 \times 50VA = 5$ kVA

kVA transferred electrically = total kVA transferred - kVA transferred magnetically

$= 15 - 5 = 10$ kVA

4. A two winding single phase transformer rated 60 kVA, 240/1200 V, 60 Hz, has an efficiency of 96% when operated at rated load, 0.8 power factor lagging. This transformer is to be utilized as a 1440/1200 step-down autotransformer in a power distribution system. (a) Assuming the transformer is ideal, that is neglect the efficiency data above for this part, determine the maximum permissible kVA rating of the autotransformer if the winding currents and voltages are not to exceed their two-winding transformer ratings given. (b) Now consider the efficiency data above, that is the equipment is not ideal anymore, determine the efficiency of the autotransformer with the kVA that you found in part (a) and 0.8 power factor leading.

(a) **Determining the maximum permissible kVA rating of the autotransformer without considering efficiency**:

The autotransformer utilizes the original transformer with the following specifications: 60 kVA, 240/1200 V.

For an autotransformer, the apparent power (kVA) is shared between the primary and secondary sides. Since the autotransformer is step-down (1440/1200), the primary side is 1440 V and the secondary side is 1200 V.

The ratio of power transmission between primary and secondary sides is determined by the voltage ratio:

Voltage Ratio = 1440/1200=1.2 V

The kVA ratio will be the square of the voltage ratio because power is proportional to the square of the voltage:

kVA Ratio = (Voltage Ratio)^2 = 1.22^2 = 1.44kVA

Now, to find the maximum permissible kVA rating on the secondary side of the autotransformer:

Maximum kVA=Rated kVA×kVA Ratio=60 kVA×1.44=86.4 kVAMaximum kVA=Rated kVA×kVA Ratio =60kVA×1.44=86.4kVA

Therefore, the maximum permissible kVA rating of the autotransformer, neglecting efficiency, would be 86.4 kVA.

(b) **Determining the efficiency of the autotransformer with the kVA found in part (a) and 0.8 power factor leading**:

$$\eta_{2winding} = \frac{60 \cdot 10^3 \cdot 1}{60 \cdot 10^3 \cdot 1 + total\ losses} = 0.96$$

Total losses = 2,5 KW

$$\boldsymbol{\eta_{AutoT/F}} = \frac{86{,}4 \cdot 10^3 \cdot 1}{86{,}4 \cdot 10^3 \cdot 1 + 2{,}5 \cdot 10^3} = \mathbf{0,9719 = 97,19\ \%}$$